

SUMMARY

BSc Artificial Intelligence student at Oxford Brookes University with interests in machine learning, algorithms, probabilistic AI and computational optimisation. Achieved a 70.3% Year 2 average, with experience developing projects across neural networks, Bayesian inference, computer vision and graph optimisation. Seeking to develop advanced theoretical and practical computer science expertise through postgraduate study.

EXPERIENCE

IT & Data Analyst

EVOLUTION CAPITAL – WORK PLACEMENT

01/2023 – 06/2023

- Conducted company research and data analysis for corporate finance activities.
- Used Microsoft Power BI to analyse and present financial data.
- Maintained and updated the firm’s website, taking responsibility for ongoing content changes.
- Skills:** Data Analysis, Power BI, Website Development

PROJECTS

Neural Traffic Routing & Congestion Optimisation

INDEPENDENT DISSERTATION PROJECT – OXFORD BROOKES UNIVERSITY

2026

- Developed a congestion-aware traffic-routing system using real-world Oxford road data, combining Dijkstra’s algorithm, A*, adaptive congestion weighting and a custom neural network for congestion prediction.
- Technologies:** Python, Neural Networks, Graph Algorithms, Dijkstra, A*, OSMnx

Bayesian Network for COVID-19 ICU Prediction

COMP5023 ADVANCED ARTIFICIAL INTELLIGENCE

2026

- Developed a Bayesian network using a real-world dataset of 5,644 patients. Implemented and compared Variable Elimination, Rejection Sampling, Likelihood Weighting and Gibbs Sampling.
- Technologies:** Python, Bayesian Networks, Probabilistic Inference, Machine Learning

AI Pathfinding & Route Optimisation

COMP5045 INTRODUCTION TO ARTIFICIAL INTELLIGENCE

2025

- Implemented and benchmarked five search algorithms on a 3,393-node Oxfordshire road network, including A*, Bidirectional MM, BFS, DFS and logic-constrained A*.
- Technologies:** Python, OSMnx, NetworkX, Matplotlib, Graph Algorithms

Eye-Controlled Morse Code Typing System

PERSONAL PROJECT

2025 – 2026

- Developed a real-time hands-free typing system using MediaPipe facial landmarks, Eye Aspect Ratio blink detection and an RBF-kernel SVM.
- Technologies:** Python, OpenCV, MediaPipe, scikit-learn, Computer Vision

PROFILES

- LINKEDIN
- GITHUB
- PORTFOLIO

SKILLS

Programming: Python, C, C++, Object-Oriented Programming

Algorithms: Data Structures & Algorithms, Graph Algorithms, Dijkstra’s Algorithm, A* Search, Algorithmic Optimisation

AI & Machine Learning: Neural Networks, Back-propagation, Gradient Descent, Bayesian Networks, Probabilistic Inference, SVM, Model Evaluation

Computer Vision: OpenCV, MediaPipe, Facial Landmark Detection, Eye Aspect Ratio
Tools: Git, OSMnx, NetworkX, Matplotlib, scikit-learn, OpenStreetMap

EDUCATION

BSc (Hons) Artificial Intelligence
2024– present
Expected classification: First-Class Honours

Year 2 average: 70.3% | Current weighted average: 67.5%
Foundations of Computation: 86%
Software Development with C/C++: 75%
Data Structures & Algorithms: 73%